11. Sequences and Series

11.11. Applications of Taylor Polynomials

11.11 Learning Objectives
- Understand how to use Taylor Polynomials to estimate functions, limits, and more.
- Understand Taylor's inequality to discuss the error when approximating functions with Taylor Series.

Example 11.11.1

How many non-zero terms of the Maclaurin series for $\cos(x)$ do you need to estimate $\cos(0.25)$ to within 10^{-16} (below machine precision)?

Solution

$$\begin{aligned}\cos(0.25) &= \sum_{n=0}^{\infty} (-1)^n \frac{(0.25)^{2n}}{(2n)!} \\ &= 1 - \frac{1}{2}(0.25)^2 + \frac{1}{4!}(0.25)^4 - \frac{1}{6!}(0.25)^6 + \ldots\end{aligned}$$

Note, this is an alternating series, so we can use the alternating series estimation theorem. The error is bounded by $\left|b_{n+1}\right|$, so we need to find n when

$$\left|(-1)^{n+1}\frac{(0.25)^{2(n+1)}}{(2(n+1))!}\right| < 10^{-16}$$

$n = 5$ has $\left|(-1)^{n+1}\frac{(0.25)^{2(n+1)}}{(2(n+1))!}\right| \approx 1.24435 \times 10^{-16}$ and $n = 6$ has $\left|(-1)^{n+1}\frac{(0.25)^{2(n+1)}}{(2(n+1))!}\right| \approx 4.27319 \times 10^{-20}$ so $n = 6$ is the stopping point we should use. Then (since we start at $n = 0$) we need seven non-zero terms to find $\cos(0.25)$ with error less than 10^{-16}.

Exercise 11.11.1

Open and experiment with the following Desmos pages to see what taking more terms in a Taylor Polynomial to approximate a function does.

https://www.desmos.com/calculator/8f41372722

https://www.desmos.com/calculator/67cc7057d1

Example 11.11.2

Prove (using series) $e^{ix} = \cos(x) + i\sin(x)$. Use this to show $e^{i\pi} + 1 = 0$.

Solution

First,

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

$$\text{so} \quad e^{ix} = \sum_{n=0}^{\infty} \frac{(ix)^n}{n!}$$

Recall: $i = \sqrt{-1}$, so $i^2 = -1$, $i^3 = -i$, $i^4 = 1$, $i^5 = i$, $i^6 = -1$ and so on. As well, since e^x is equal to it's Taylor series everywhere (check $R_n(x) \to 0$) we can rearrange the terms. So,

$$\begin{aligned}
e^{ix} &= \sum_{n=0}^{\infty} \frac{(ix)^n}{n!} \\
&= 1 + (ix) + \frac{(ix)^2}{2} + \frac{(ix)^3}{3!} + \frac{(ix)^4}{4!} + \frac{(ix)^5}{5!} + \frac{(ix)^6}{6!} + \ldots \\
&= 1 + ix + \frac{i^2x^2}{2} + \frac{i^3x^3}{3!} + \frac{i^4x^4}{4!} + \frac{i^5x^5}{5!} + \frac{i^6x^6}{6!} + \ldots \\
&= 1 + ix - \frac{x^2}{2} - \frac{ix^3}{3!} + \frac{x^4}{4!} + \frac{ix^5}{5!} - \frac{x^6}{6!} + \ldots \\
&= 1 - \frac{x^2}{2} + \frac{x^4}{4!} - \frac{x^6}{6!} + \ldots + ix - \frac{ix^3}{3!} + \frac{ix^5}{5!} + \ldots \\
&= 1 - \frac{x^2}{2} + \frac{x^4}{4!} - \frac{x^6}{6!} + \ldots + i\left(x - \frac{x^3}{3!} + \frac{x^5}{5!} + \ldots\right) \\
&= \cos(x) + i\sin(x)
\end{aligned}$$

So, $e^{i\pi} = \cos(\pi) + i\sin(\pi) = -1 + 0$, thus $e^{i\pi} + 1 = -1 + 1 = 0$.

11.11b Section Summary:

- We used Maclaurin series to approximate function values.
- We used series to prove the important relation $e^{i\pi} + 1 = 0$.